

# Jude Lwin

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## Education

### University of Maryland, College Park

GPA: 3.9/4.0

*Bachelor of Science, Computer Science*

*May 2027*

- Coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Distributed Systems, Functional Programming, Machine Learning, Data Science

## Experience

### Microsoft

Redmond, WA

*Incoming Software Engineer Intern*

*May 2026 – Aug. 2026*

### Amazon

New York, NY

*Software Engineer Intern*

*Feb. 2026 – Apr. 2026*

- Extended authorization infrastructure (41K+ applications, 198K+ service relationships) to secure **agentic AI** by building a **Python hook**-based runtime enforcement layer to prevent unsafe behavior caused by prompt injection
- Designed a deterministic **policy engine** to track cumulative risk and block unsafe multi-step tool combinations
- Built **Java** backend on **AWS Lambda** exposing 12 tools via **MCP**, integrating with **OAuth**-based authorization
- Executed end-to-end attack simulations across 4 scenarios on an onboarded agent, demonstrating human-in-the-loop approval for unsafe actions and full **MCP**-to-auth execution flow

### Capital One

McLean, VA

*Software Engineer Intern*

*Jun. 2025 – Aug. 2025*

- Built LLM evaluation pipeline for **RAG** chatbot in **Python/Snowflake**, scaling model validation from dozens to 10K+ queries per deployment, reducing engineering time from days of manual review to overnight batch processing
- Trained ensemble classifiers with **scikit-learn** (Random Forest, XGBoost) and developed **NLP** scoring system (ROUGE, Levenshtein, fuzzy encoding) achieving 91.2% accuracy in deployment verification
- Engineered backtesting framework to validate metric/classifier updates against all previous evaluation runs

### University of Maryland

College Park, MD

*Research Engineer | Advisor: Dr. Furong Huang, Dr. Zikui Cai*

*Sep. 2025 – Present*

- Engineered **Python** infrastructure for training a **VLM** to edit PowerPoint slides, expanding **Zenodo** PPTX corpus from 10,448 to 18,044 files (242GB → 483.5GB) via resumable **REST API** crawler with **fcntl** locking
- Designed **distributed** triplet pipeline on **Nexus HPC**, converting decks into **PPTX/schema/PNG** training examples across 16 **Slurm** array tasks with **sharding** and resumable execution
- Developed **PPTX-to-schema** tools for VLM slide editing with 46% token reduction and 95.1% round-trip fidelity
- Architected **RAG** pipeline for math reasoning using reasoning-trace retrieval over DeepMath-103K (FAISS), improving AIME accuracy from 4.5% to 41.9% on DeepSeek-R1; **ACL** SRW submission

### WISE Cities

College Park, MD

*Software Engineer Intern | AI/ML Club*

*Feb. 2025 – May 2025*

- Led backend development of a recommendation system for older adults to find local organizations and activities
- Engineered scalable **gRPC** microservice with **MongoDB** querying and **Pydantic** data validation
- Integrated all-MiniLM-L6-v2 and **Pinecone** to enable sub-100ms k-NN recommendations

### University of Maryland

College Park, MD

*Teaching Assistant, CMSC330 (Programming Languages)*

*Jan. 2025 – Present*

- Led weekly discussion for 30+ students, hosted 5+ weekly office hours, and coordinated grading with 39 TAs
- Maintained 7 **OCaml/Rust** projects with 90+ tests for 900+ students; deployed via GitHub Actions and **Docker**

## Projects

### Distributed Build System 🌀 | C++, gRPC, Redis, Docker

- Built **C++** build system for parallelized compilation using **gRPC** coordination and dependency-graph scheduling
- Added artifact caching with **Redis** and containerized benchmarks to evaluate scalability and fault recovery

### Smart Study Assistant 🌀 | Python, FastAPI, Celery, Docker, Supabase, Pinecone, AWS S3, React

- Developed distributed study assistant for cited course-specific search with **FastAPI**, **React**, and **Docker**
- Implemented RAG pipeline (**Celery**, **Pinecone**) for PDF parsing, chunking, and sub-200ms semantic search

## Technical Skills

**Languages:** Python, C++, Java, TypeScript, JavaScript, Rust, Go, C, OCaml, SQL, R

**Frameworks/Libraries:** PyTorch, Scikit-Learn, Pandas, FastAPI, React, Node.js, Express, Flask, Pydantic

**Technologies/Tools:** Git, AWS, Docker, Linux, PostgreSQL, MongoDB, Snowflake, Redis, gRPC, REST APIs, CI/CD